

Project Smart Trigger for ECE/CS 3992

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Abstract—Some people will read this and nothing else. What should they know about your proposal?

Index Terms—IEEE, IEEEtran, journal, L^AT_EX, paper, template.

I. PROBLEM STATEMENT

What is the problem the project is trying to solve?

II. SOLUTION SPACE

Efforts to improve firearm safety through access control have evolved through several distinct technological approaches. At the most basic level, traditional solutions rely on purely mechanical safeguards such as trigger locks or integrated key-based locking mechanisms. These systems are inexpensive and highly reliable because they do not depend on electronics or power, but they fundamentally lack user-specific authentication. Anyone with access to the key or combination can operate the firearm, which causes concern toward illegitimate use. In time-sensitive or high-stress situations, the need to manually unlock a device with a key can render the safety ineffective in practice, leading many police and military personnel to avoid using such systems altogether. This creates a gap between theoretical safety and real-world adoption, motivating the development of systems that can enforce access control without requiring manual, non-instantaneous intervention.

Electronic authentication methods represent the next step in this progression. Systems based on PIN entry, external key fobs, or smartphone connectivity attempt to introduce user-specific control while maintaining relatively simple hardware implementations. While these approaches improve upon mechanical systems by enabling programmable access, they still require explicit interaction from the user, such as entering a code or activating a device. This introduces latency and cognitive load that can be unacceptable in defensive scenarios. Additionally, wireless-based systems may be vulnerable to interception or spoofing, depending on their implementation. As a result, these approaches highlight the need for authentication mechanisms that are both secure

and implicit, allowing the system to verify the user seamlessly without requiring conscious action.

Biometric authentication has emerged as the most prominent alternative in this space, particularly in commercial “smart gun” designs. By leveraging physiological traits such as fingerprints, palm vein patterns, or even grip characteristics, biometric systems aim to bind firearm operation directly to the identity of the user. This approach is compelling because it eliminates the need for external tokens and ensures that only authorized individuals can operate the weapon. However, when examined in the context of real-world firearm use, biometrics introduce a number of critical challenges. Sensors such as fingerprint scanners are highly sensitive to environmental conditions. Metrics like dirt, sweat, gloves, or moisture can significantly degrade performance. Firearms are often used in unpredictable and harsh environments, where such conditions are common, leading to an increased likelihood of authentication failure.

Latency is another major concern. Biometric systems require time to capture, process, and match data against stored templates. Even small delays can be unacceptable in scenarios where immediate response is necessary. Equally important is the issue of reliability in terms of false negatives and false positives. A false negative, where an authorized user is denied access, represents a potentially dangerous failure in a defensive situation. Conversely, a false positive undermines the entire purpose of the system. These risks are compounded by the long-term degradation of sensors and the variability of human biological signals. In addition to these technical concerns, biometric systems tend to be significantly more expensive and power-intensive than alternative solutions. Commercial implementations often cost on the order of \$1000 to \$1500, reflecting the complexity of most biometric devices. This cost and power burden conflicts directly with the constraints of a small, battery-powered module designed to fit within the firing control group of an AR-15.

RFID-based authentication offers a different point in the design space by prioritizing simplicity, robustness, and low power consumption. In a passive near-field

RFID system, authentication occurs when a tag enters the electromagnetic field generated by a reader, allowing the tag to transmit its unique identifier without requiring its own power source. This enables extremely low-latency operation and reduces the system's overall energy requirements. RFID systems are also less sensitive to environmental contaminants compared to optical or capacitive biometric sensors, making them more reliable in varied conditions. The primary tradeoff is that RFID relies on possession rather than inherent identity. A user must carry an external token, such as a ring or card, which introduces the possibility of loss or unauthorized use if the token is compromised.

More complex hybrid systems attempt to combine multiple authentication methods, such as pairing RFID with biometrics or behavioral sensing, in order to improve security. While these approaches can reduce certain vulnerabilities, they significantly increase system complexity, cost, and power consumption. For a compact, retrofittable module operating under strict physical and electrical constraints, this hybrid complexity is impractical. As a result, the solution space reflects a series of tradeoffs between security, reliability, usability, and feasibility. Biometric systems offer strong identity assurance but struggle with robustness and cost, while RFID systems sacrifice some degree of identity binding in exchange for speed, reliability, and practicality. This balance makes RFID a compelling candidate for a system intended to be compact, cost-effective, and deployable in real-world conditions.

III. RELATED WORK

Radio frequency identification (RFID) has been widely studied as part of the broader class of automatic identification technologies. Kaur et al. provides a comprehensive overview of its principles, capabilities, and limitations. RFID systems enable the wireless transmission of identification using radio waves, allowing individuals or objects to be identified without requiring line-of-sight interaction, in contrast to traditional barcode systems [1]. This capability significantly improves automation and reduces the need for human action within lock systems, making RFID useful for embedded and access control applications that require high speed and reliability.

A major focus of prior work is the distinction between passive and active RFID systems, as well as the physical mechanisms that enable their operation. Passive RFID operates by harvesting energy from the electromagnetic field generated by a reader. In near-field systems, this process is governed by magnetic induction, where an alternating magnetic field induces a voltage across a

tag's coil, which provides enough energy for the tag to transmit its stored ID [1]. This eliminates the need for an onboard power source. These characteristics make passive RFID well-suited for compact, power-constrained embedded systems, such as our intended implementation within a firearm's firing control group.

Despite these advantages, Kaur et al. also identifies several important limitations of RFID systems that must be addressed. One key constraint is the relationship between operating frequency and read range. In near-field systems, the effective communication range is limited and decreases rapidly due to the inverse cubic decay of the magnetic field [1]. While this restricts long-range communication, it can be advantageous for security sensitive applications by ensuring that the authenticated user is within close proximity. Other challenges include signal collision when multiple tags are present, environmental interference caused by conductive materials or electromagnetic noise, and the lack of universal standardization across RFID systems [1]. Security and privacy concerns are also significant, as unauthorized cloning of tags can compromise system integrity if appropriate safeguards are not implemented.

Taken together, the existing literature establishes RFID as a mature and well-understood technology for identification and access control. It has demonstrated advantages in automation, robustness, and low power operation. At the same time, it highlights key technical and security challenges that must be carefully addressed in our implementation. This paper provides a strong foundation for the proposed system, informing both the design choices and the tradeoffs involved in applying RFID to firearm safety.

IV. SOLUTION / CONTRIBUTION

To truly understand our project and our approach, an overview of the firing control group of an AR-15 rifle is necessary. The firing control group operates through a precise mechanical interaction between the trigger, sear, and hammer. As seen in Figure 1,

the hammer is held back under spring tension by the sear. When the trigger is pulled, it disengages the sear, releasing the hammer and allowing it to rotate forward where it would eventually hit the firing pin. Our proposed solution is to block the firing control group by applying an additional force against the hammer, ensuring that it can never strike the firing pin. Our idea is to use a linear actuator to provide this force against the hammer. A linear actuator paired with a microcontroller would allow for low-voltage operation of our access control. The gun would then only be able to fire when a validated RFID

tag is present in close proximity to our implemented system on the firing control group.

The primary contribution of this project is the integration of RFID-based access control into this already operating, small mechanical system. A passive RFID reader embedded near the control group continuously scans for a valid tag worn by the user, particularly in a ring worn on their finger. When an authorized tag is detected, the system signals the actuator to disengage, allowing normal operation of the firing mechanism. If no valid tag is present, the actuator blocks the hammer from hitting the firing pin. This way the firearm would never fire even if the trigger is pressed. This approach enables fast and implicit authentication without requiring user interaction, while also remaining low-power and cost-effective compared to biometrics alternatives. By combining a simple mechanical intervention with proximity-based RFID authentication, the system provides a practical solution for enhancing firearm safety.

V. SAFETY, GLOBAL, SOCIETAL, ETHICAL CONSIDERATIONS

What are the public health, safety and welfare, global, cultural, societal, environmental, and economic factors related to the problem and solution space? What power does the proposed system provide, and to whom? What are the relevant professional ethical obligations to consider in the design of the system?

A. Factors Related to the Problem and Solution

- 1) *Public Health:*
- 2) *Safety and Welfare:*
- 3) *Global:*
- 4) *Cultural:*
- 5) *Societal:*
- 6) *Environmental:*
- 7) *Economic:*

B. Power

VI. PROPOSED WORK

A project includes some things and excludes others you know you won't work on. Define more explicitly what is included and not included here. What will be demonstrated at project completion at Demo Day?

VII. PROJECT TASKS

Describe each task.

A. Major Milestones

- Planning:
 - Full bill of materials made
 - Major research complete
 - Safety precautions addressed
 - Due:
- Functional RFID reader:
 - Able to read and log tags
 - Exportable log
 - Due:
- Application for updating authorized tags:
 - User able to update authorized tags without reflashing
 - Due:
- Successful blocking of fire control:
 - Method of blocking the firing of the AR-15 solidified and location confirmed
 - Due:
- Part fitting:
 - Adapting all components into the small form factor required by constraints
 - Due:
- Assembly:
 - All components fit together and operate in basic expected function
 - Due:
- Endurance testing:
 - Full system able to withstand operation of firearm
 - Due:

B. Stretch goals

things that you are planning to include, but the project will succeed even without them (and why)

C. Purchase, build, borrow

What components will be purchased and what will be built? What resources are needed to complete the tasks? These may be your own, or in a lab, or may be people who will help you.

D. Task interfaces

What are the important inputs and outputs of each individual component? Define them.

E. Testing

How will each task be tested

F. Design approach

For each component, how will it be designed? What tools will be used (software, hardware, measurement, etc.)

G. Current status

What tasks have been finished? What is left to be done?

H. Parts

Complete parts list with vendor name and information

VIII. INTEGRATION STRATEGY

Our project has a constraint to fit within a very small space. We will utilize a partially distributed integration strategy in order to mesh all of the components together. The primary electronics are an RFID module, a microcontroller unit (MCU), a power supply, and a linear actuating solenoid. All components except the solenoid will be housed in the plastic grip for an AR-15. The plastic grip has the benefit of not interfering with the RFID signal thus reducing the complexity of a faraday cage effect that could block any electromagnetic signal from being identified.

The solenoid will be housed inside the actual receiver of the AR-15. This separate location allows for the direct blocking of the fire control group. The actuation will be triggered via fine gauge wire passing through one of the holes where the grip physically attaches to the receiver.

Power management is one of the most challenging parts of this entire project. The RFID module can consume up to 60mA during its operation and the solenoid up to 200mA. To compensate for these power requirements we will employ an interrupt driven state machine. A low power timer will periodically activate the RFID module to check for a tag, if one is detected it will activate hardware interrupt on the MCU to check if the tag is authorized. If the tag is in the authorized list the MCU will then actuate the solenoid to remove the block of the fire control group, ultimately allowing the firing of the AR-15.

IX. RISK ASSESSMENT

What are the largest risks to successful demonstration of the project? Describe each risk as green, yellow, or red. At proposal stage, your goal is to have all risks in the "green" category, that is, you have de-risked any major risks to the project. Describe the de-risking process.

X. CONCLUSION

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